

Triangle sums, peripheral cycles, and contractions for orderable cographic matroids

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Abstract

For a finite simplicial triangulation T of a closed surface, let Λ_T be the graph on the edges of the 1-skeleton in which two edges are adjacent when they lie in a common triangular face. We study the property $U(T)$ that $\Lambda_T[B]$ is connected for every bond B of the 1-skeleton. The induced graph is automatically 2-regular, so $U(T)$ provides a consistent ordering of the circuits of the cographic matroid.

We first prove a graph-theoretic gluing theorem for cographic matroids with a compatible rooted triangle. In the surface setting this becomes an exact connected-sum theorem:

$$U(T_1 \#_{\Delta} T_2) \iff U(T_1) \text{ and } U(T_2).$$

Together with the projective-plane theorem proved in an earlier paper, this gives infinitely many 3-connected, regular, non-graphic, orderable cographic matroids on every closed nonorientable surface. We then establish two complementary structural restrictions. A nonfacial induced cycle whose vertex deletion leaves the graph connected must be one-sided, and $U(T)$ descends under every legal edge contraction. Consequently, existence on a fixed surface reduces to its finite set of irreducible triangulations. We also give an explicit Klein-bottle example showing that U is not preserved by general vertex splitting. These results provide composition, obstruction, and reduction mechanisms without claiming a classification of all orderable cographic matroids or all surface triangulations satisfying U .

1 Introduction

Crenshaw and Oxley introduced *orderable matroids*: matroids whose circuits admit reversible cyclic orderings that agree on every shared adjacent pair [1]. Graphic matroids are naturally orderable, while the binary case is structurally difficult. Their current author manuscript proves that every 4-connected regular orderable matroid is graphic [2, Theorem 1.5]. The exact 3-connected boundary is different. An earlier paper [3] proved that every finite simplicial triangulation of the real projective plane gives an orderable cographic matroid, yielding infinitely many 3-connected regular non-graphic examples.

The earlier proof used a particular global adjacency relation: two graph edges are adjacent when they meet in a triangular face. Restricted to any bond, this relation is a disjoint union of cycles, and a topological argument on the projective plane forces a single cycle. The present

paper asks which operations preserve this stronger, specified ordering and what its failure reveals about the topology of a triangulation.

The results have three layers. First, Section 3 gives a graph-theoretic triangle-rooted gluing theorem that is not restricted to surface triangulations. Second, Section 4 proves that the surface property is preserved and reflected by simplicial connected sum along a triangular face. This extends [3] from the projective plane to explicit infinite families on every closed nonorientable surface. Third, Sections 5 and 6 give a topological obstruction and a contraction theorem. The obstruction limits peripheral cycles in any positive example; the contraction theorem reduces the existence question on each fixed surface to finitely many irreducible triangulations.

The specified property studied here is stronger than abstract orderability: $U(T)$ implies that $M^*(T^{(1)})$ is orderable, but an abstract consistent ordering need not arise from the triangular faces of the given embedding. We keep this distinction throughout.

2 Definitions and the face-adjacency property

All graphs and simplicial complexes are finite. If G is connected, the circuits of its cographic matroid $M^*(G)$ are the bonds of G . Equivalently, for a nonempty proper set $S \subset V(G)$, the cut $\delta_G(S)$ is a bond exactly when both $G[S]$ and $G[V(G) \setminus S]$ are connected.

We use the following sufficient form of consistency.

Lemma 2.1 (global adjacency). Let M be a matroid. Suppose that a simple graph A on $E(M)$ has the property that $A[C]$ is a cycle for every circuit C of M . Reading each cycle in either direction gives a consistent family of reversible cyclic orderings. Hence M is orderable.

Proof. Whether two elements are adjacent is determined by the single global graph A . Thus, whenever two circuits contain the same pair, that pair is adjacent in one induced ordering if and only if it is adjacent in the other. $\square$

Let T be a simplicial triangulation of a closed connected surface, let $G = T^{(1)}$, and define the *face-adjacency graph* Λ_T on $E(G)$ by joining two distinct graph edges when they belong to a common triangular face. For a cut B , each triangular face contains either zero or two edges of B . Moreover, each cut edge lies in two faces. It follows that $\Lambda_T[B]$ is 2-regular.

We say that T has the *single-boundary property*, denoted by $U(T)$, when $\Lambda_T[B]$ is connected for every bond B of G . Thus $U(T)$ is equivalent to the statement that this particular face-adjacency relation induces a cycle on every bond. By Lemma 2.1,

$$U(T) \implies M^*(T^{(1)}) \text{ is orderable.}$$

There is also a useful topological interpretation. Give the vertices on the two sides of a cut the values 0 and 1, extend affinely over each simplex, and take the level set at $1/2$. Its segments through the mixed faces form a disjoint union of circles, and their combinatorial adjacency graph is exactly $\Lambda_T[B]$. Thus $U(T)$ says that the level set associated with every bond has one component.

3 A triangle-rooted gluing theorem

The first result is independent of surface topology.

Let H be a graph containing a triangle R with edges r_1, r_2, r_3 . Suppose $M^*(H)$ has a consistent ordering represented by a global adjacency graph A_H as in Lemma 2.1. We call this ordering $*$ compatible with the rooted triangle R^* when

$$A_H[\{r_1, r_2, r_3\}] = K_3.$$

This does not require R to be a bond or a face in a specified embedding.

Theorem 3.1 (triangle-rooted gluing). Let H_1 and H_2 be finite simple 3-connected graphs whose intersection is exactly a common triangle R :

$$V(H_1) \cap V(H_2) = V(R), \quad E(H_1) \cap E(H_2) = E(R).$$

Assume that each $M^*(H_i)$ has a global consistent adjacency graph A_i compatible with R . If $G = H_1 \cup_R H_2$, with the three edges of R retained, then $M^*(G)$ is orderable.

Proof. Form a graph A on $E(G)$ by taking $A_1 \cup A_2$ and deleting the three adjacencies between pairs of root edges. We prove that $A[B]$ is a cycle for every bond B of G .

Use the two sides of B to color $V(G)$ with two colors. If the three vertices of R are monochromatic, the opposite color cannot occur in the interiors of both factors, since those vertices would have no same-colored route between the factors. In the active factor, every component of the root-colored side meets R , and the monochromatic triangle R joins all such components. Thus B lies entirely in one factor, where it is a bond and contains no root edge. The deletion of root-root adjacencies does not affect it, so $A[B] = A_i[B]$ is a cycle.

Suppose instead that R is bichromatic. Exactly two root edges, say r_a and r_b , cross the cut. In either factor, every component induced by one color must meet R , since otherwise it could not connect through the other factor. The same-colored vertices of R are adjacent within the triangle, so each color induces a connected subgraph in each factor. Therefore $B_i = B \cap E(H_i)$ is a bond of H_i and contains r_a, r_b .

Compatibility makes $r_a r_b$ an edge of the cycle $A_i[B_i]$. Deleting that adjacency leaves a Hamilton path P_i from r_a to r_b . Since a 3-connected graph has no bond of size two, each path contains a non-root element. The paths have only their endpoints in common, and $P_1 \cup P_2 = A[B]$ is a cycle through every element of B . Lemma 2.1 now proves the result. $\square$

If H_i is the 1-skeleton of a closed surface triangulation satisfying U , the face-adjacency graph supplies such a rooted ordering at every triangular face. The theorem also applies beyond triangulations. For example, a 3-connected plane graph with a triangular face has a compatible cographic ordering coming from its planar dual and can be glued to any other compatible factor.

4 Simplicial triangle sums

Let T_1, T_2 be closed connected simplicial triangulations, with chosen triangular faces F_1, F_2 . Delete the interiors of those faces and identify their boundary triangles by a vertex bijection. Assume that the interiors of the two complexes are otherwise disjoint. Write the resulting triangulation as $T_1 \#_{\Delta} T_2$. The identified triangle is retained in the 1-skeleton but is no longer a face; we call it the *seam*.

Theorem 4.1 (triangle-sum equivalence).

$$U(T_1 \#_{\Delta} T_2) \iff U(T_1) \text{ and } U(T_2).$$

Proof. The forward construction in Theorem 3.1 applies to the face-adjacency graphs of the factors. Equivalently, consider a bond in the sum. If the seam vertices are monochromatic, the bond lies in one factor. If they are bichromatic, its restriction to either factor is a bond containing the same two seam edges. In each capped factor the corresponding boundary is one circle. Removing the chosen face turns that circle into an arc between the midpoints of the two crossing seam edges, and the two arcs glue to one circle. Hence $U(T_1)$ and $U(T_2)$ imply $U(T_1 \#_{\Delta} T_2)$.

For the converse, suppose that T_1 fails U . Choose a bond of its 1-skeleton whose face-adjacency boundary has $k \geq 2$ components and use its sides to color the vertices. If the chosen face is monochromatic, give all non-seam vertices of T_2 that same color. The cut remains a bond in the sum and retains all k boundary components.

If the chosen face is bichromatic, give every non-seam vertex of T_2 the majority color on the seam. The minority-colored side in T_2 is a single seam vertex. The other side is connected because the 1-skeleton of a closed simplicial triangulation is 3-connected and remains connected after that vertex is deleted. Thus the extended cut is again a bond. In the punctured T_2 , the boundary around the minority seam vertex is one arc. It glues only to the component of the original boundary that crossed the deleted face; the remaining $k - 1$ circles persist. The sum therefore fails U . The same argument applies with the factors reversed. $\square$

The equivalence iterates along any tree of triangle sums. In particular, a triangulation decomposed along separating nonfacial triangles satisfies U if and only if every capped leaf factor does. The statement does not require uniqueness of the chosen decomposition tree.

Corollary 4.2 (triangular disk replacement). Let D be a simplicial triangulated disk with triangular boundary, whose interior vertices and edges are disjoint from a closed triangulation T . Replacing a triangular face of T by D preserves and reflects U .

Proof. If D is one triangle there is nothing to prove. Otherwise, cap D with a new triangular face. The result is a sphere triangulation, which satisfies U : in the planar dual, every bond of the primal graph corresponds to one dual cycle, with exactly the face-adjacency order. Apply Theorem 4.1. $\square$

4.1 Infinite families in every nonorientable genus

Let N_h denote the closed nonorientable surface with h crosscaps. The earlier projective-plane theorem [3] says that every finite simplicial triangulation of $N_1 = \mathbb{RP}^2$ satisfies U .

Theorem 4.3. For every integer $h \geq 1$, take any h finite simplicial triangulations of $\mathbb{RP}^2$ and join them successively along triangular faces. The resulting triangulation T of N_h satisfies $U(T)$. For every h , this construction yields infinitely many pairwise nonisomorphic 3-connected, regular, non-graphic, orderable cographic matroids.

Proof. The first assertion follows by repeated application of Theorem 4.1. The 1-skeleton of a closed simplicial triangulation is 3-connected. It can also be seen directly here: after deleting at most two vertices, each factor remains connected and every seam retains a vertex joining its two sides.

If the triangulation has n vertices and m edges, Euler's formula gives

$$m = 3n - 6 + 3h > 3n - 6.$$

Its 1-skeleton is therefore nonplanar. Its cographic matroid is regular and binary, is 3-connected, and is non-graphic by the graphic-cographic planarity criterion for 3-connected

graphs. It is orderable by $U(T)$ and Lemma 2.1. Finally, repeated stellar subdivision of a triangular face is a special case of Corollary 4.2 and increases the ground-set size, giving infinitely many pairwise nonisomorphic matroids. $\square$

5 A peripheral-cycle obstruction and a failed inverse operation

The connected-sum theorem supplies positive constructions. The next result is a necessary condition for an arbitrary triangulation to satisfy U .

Theorem 5.1 (peripheral-cycle obstruction). Let T be a closed connected simplicial triangulation satisfying $U(T)$, with $G = T^{(1)}$. If C is an induced cycle of G , $G - V(C)$ is connected, and C is not a triangular face of T , then C is one-sided in the surface.

Proof. Since $G[V(C)] = C$ and $G - V(C)$ are connected, $\delta_G(V(C))$ is a bond. Because C is induced and nonfacial, the full subcomplex of T induced by $V(C)$ is the circle C , rather than a disk. In every simplex, the lower half of the two-valued piecewise-linear function associated with the cut collapses linearly onto its face spanned by the vertices of C . These simplexwise collapses agree on common faces. The standard derived-neighborhood criterion therefore identifies the lower half as a PL regular neighborhood of the full induced subcomplex C [6]. Its boundary components are precisely the components of $\Lambda_T[\delta_G(V(C))]$.

If C is two-sided, its regular neighborhood is an annulus and has two boundary components, contradicting $U(T)$. A one-sided cycle has a Möbius-band neighborhood and one boundary component, so it is the only remaining possibility. $\square$

Corollary 5.2. Every two-sided nonfacial induced cycle in a triangulation satisfying U is vertex-separating. In particular, every two-sided nonfacial triangle is a 3-vertex separator.

The inverse of edge contraction is vertex splitting. It is natural to ask whether arbitrary legal vertex splits preserve U , which would turn finite irreducible triangulations into generators for all positive examples. The answer is no.

Proposition 5.3. The property U is not preserved by general legal vertex splitting, even for 3-connected triangulations of the Klein bottle.

Proof. Begin with the following nine-vertex Klein-bottle triangulation, obtained as the triangle sum of two six-vertex projective-plane triangulations:

```
014 017 024 025 035 038 067 068 123
125 136 145 168 178 234 345 367 378
```

It satisfies U by Theorem 4.1. The link of vertex 0 in cyclic order is

```
1 4 2 5 3 8 6 7.
```

Split the two link arcs at positions 1 and 5, introducing vertex 9. The resulting faces are

```
014 017 049 067 068 089 123 125 136 145
168 178 234 249 259 345 359 367 378 389.
```

This is a closed 3-connected Klein-bottle triangulation and has no separating nonfacial triangle. For $S = \{0, 3, 4, 6\}$, the cut $B = \delta(S)$ is a bond. Its 17 edges are partitioned by face adjacency into the following two cycles:

01 07 67 37 38 39 35 45 14
08 09 49 24 23 13 16 68

Thus $\Lambda_T[B]$ is disconnected and the split triangulation fails U . $\square$

An exact exhaustive check of all 90 nontrivial single splits of the same labelled base triangulation gives 63 positive operations retaining a separating nonfacial triangle and 27 negative operations with no such triangle. These operations produce 78 distinct labelled face sets: 52 positive and 26 negative. This finite table is a calibration around one base example, not a classification up to isomorphism and not a general equivalence between seams and U .

6 Descent under legal edge contraction

Although U need not survive the inverse operation, it always descends under legal edge contraction.

Let $e = uv$ be an edge of a closed simplicial triangulation T satisfying the usual link condition, so that contracting e gives another simplicial triangulation $T' = T/e$ of the same surface. Let w be the image of u, v .

Theorem 6.1 (contraction descent).

$$U(T) \implies U(T/e).$$

Proof. Let $B' = \delta_{T'(1)}(S')$ be a bond and color its two sides. Lift the coloring to T by giving both u and v the color of w . The resulting cut B is a bond. Indeed, a monochromatic path in T' lifts directly except possibly when it enters w through an edge incident only with u and leaves through an edge incident only with v ; in that case the monochromatic edge uv joins the two lifted pieces. Hence both color classes remain connected.

By the link condition, the common neighbors of u and v are exactly the third vertices a, b of the two faces incident with uv . Away from uv , the only edge identifications under contraction are

$$ua, va \mapsto wa, \quad ub, vb \mapsto wb.$$

The full link condition also prevents two distinct surviving faces from colliding under these identifications. Thus, after the two incident faces uva and uvb disappear, every other face of T corresponds uniquely to a face of T' .

For $z \in \{a, b\}$ of the same color as u, v , neither uz nor vz belongs to B , so no cut-edge identification occurs. If z has the opposite color, then $uz, vz \in B$ and those two edges are adjacent in $\Lambda_T[B]$ through the disappearing mixed face uvz ; contraction replaces them by the single edge wz . All other mixed faces and their induced adjacencies correspond bijectively before and after contraction. Consequently $\Lambda_{T'}[B']$ is obtained from the connected graph $\Lambda_T[B]$ by contracting zero, one, or two of these specified adjacency edges. It is therefore connected. As in Section 2 it is also 2-regular, so it is a single cycle. Since B' was arbitrary, $U(T')$ holds. $\square$

A triangulation is *irreducible* if no edge can be legally contracted while remaining a triangulation of the same surface.

Corollary 6.2 (finite irreducible reduction). Every closed surface triangulation satisfying U contracts through U -preserving steps to an irreducible triangulation of the same surface.

Hence a fixed closed surface has a triangulation satisfying U if and only if one of its irreducible triangulations satisfies U .

Proof. Repeatedly apply Theorem 6.1 until no legal contraction remains. The reverse implication in the existence statement takes the irreducible triangulation itself. Each fixed closed surface has only finitely many irreducible triangulations [5]. $\square$

The corollary is an existence reduction, not a classification of all positive triangulations. Proposition 5.3 shows exactly why the reduction cannot simply be reversed by allowing arbitrary vertex splits. Complete irreducible lists are known for several low-genus surfaces [4], but those enumerations are not used in the proof.

7 Verification and scope

The source package contains three separate finite checks. They are independent of the general proofs but share a common exact implementation of graph bonds and face adjacency:

1. they build iterated triangle sums of the six-vertex projective-plane triangulation and exhaust all bonds through nonorientable genus five;
2. they reconstruct Proposition 5.3, check the topology and 3-connectivity of every split result, and exhaust every bond in all 90 split operations;
3. they check a bounded collection of two-step splits as a diagnostic for a possible stronger seam theorem.

The corresponding entry points are `verify_nonorientable_triangle_sum_family.py`, `check_vertex_split_nonclosure.py`, and `probe_two_step_seam_persistence.py`. The first two support statements recorded in this manuscript. The third returned no positive prime example in its bounded search, but that `NO_HIT` is not used as evidence for an existence, nonexistence, or classification theorem. The proofs of Theorems 3.1, 4.1, 4.3, 5.1, and 6.1 are general and do not depend on the finite checks. The release artifact will record the exact commands, shared dependency, and file hashes.

We make no claim that every orderable cographic matroid arises from a surface triangulation, that every abstract ordering is a face-adjacency ordering, or that the prime triangulations satisfying U have been classified. We also do not claim that the rooted compatibility condition in Theorem 3.1 is necessary. Every constructed example contains triangular cycles, which give three-element cocircuits and exact 3-separations; for $h \geq 2$, the seam triangles give additional such separations. The examples therefore do not contradict the 4-connected result of Crenshaw and Oxley [2].

The current literature check covered the published paper [1], the latest author manuscript [2], the preceding projective-plane construction [3], and the standard irreducible-triangulation sources [4, 5]. No direct statement of the triangle-rooted gluing theorem, the triangle-sum equivalence, the peripheral obstruction, or the contraction-descent theorem was located. This records the search boundary; it is not a claim of exhaustive worldwide priority. The present manuscript is an internally developed candidate and has not undergone external peer review.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use, and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.

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